Aya Anwar

Full Stack Engineer | Node.js · NestJS · React · Angular

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Summary

Full Stack Engineer who ships end-to-end web apps: RESTful APIs with Node.js, NestJS, and Express on MongoDB and SQL databases, and responsive frontends in React and Angular. Backend lead of a 5-member team on a multi-role graduation project management platform (JWT auth, RBAC, real-time chat, AI chatbot), and sole full stack developer of an e-commerce platform that serves real customers. Adding DevOps depth with Linux, Docker, Kubernetes, Ansible, Nginx, and AWS EC2. Seeking a Junior Full Stack Engineer or Internship role.

Projects

Graduate Project Management Platform | 12/2025 – 05/2026

NestJS, TypeScript, MongoDB (Mongoose), JWT, Passport, Cloudinary, React 19, Vite, Tailwind CSS, TanStack Query | GitHub | Live

- Architected the NestJS backend for a platform that manages graduation projects for universities, serving 3 user roles (student, supervisor, admin) as the backend owner in a 5-member team.
- Implemented JWT authentication (Passport, Bcrypt, cookie-based sessions) and role-based access control (RBAC) to restrict every endpoint by role.
- Integrated real-time student–supervisor chat and an AI chatbot (n8n) that answers student questions on the platform, and connected the API to the React 19 + TypeScript client (TanStack Query, Axios, multi-language support with i18next).
- Handled file uploads with Multer and Cloudinary, email notifications with Nodemailer, and centralized logging with Winston.
- Enforced input validation with class-validator and organized the code into modular, clean-architecture layers so multiple universities can adopt the platform.

DevOps Project – AWS EC2 Web Application Deployment | 01/2026 – 02/2026

AWS EC2, EBS, Linux, MariaDB, Python/Go | GitHub

- Deployed a production-ready note-taking application on an AWS EC2 instance with a Python/Go backend and MariaDB for persistent storage.
- Configured the Linux server and security groups to expose only SSH and HTTP access.
- Automated EBS-backed database backups to protect data against instance failure.
- Delivered secure CRUD functionality covering cloud provisioning, infrastructure setup, and backup recovery.

Online Tutoring Platform | 09/2025 – 12/2025

MongoDB, Express.js, Angular, Node.js | Live

- Created a booking system that lets students reserve study stages through interactive forms.
- Engineered admin controls to manage bookings, stages, and lesson access.
- Modeled a stages → units → lessons hierarchy that supports video, PDF, and image content.
- Added quizzes per lesson and unit to track student progress, plus discussion sections for each stage.

ATHAR – Fashion E-Commerce Platform | 07/2025 – 10/2025

MongoDB, Express.js, Angular, Node.js | Live

- Launched a production e-commerce platform for the ATHAR brand as the sole developer; it currently serves real customers and orders.
- Implemented secure RESTful APIs for product categories, brand management, inventory tracking, cart and order flow, multi-address checkout, and customer feedback.
- Owned the full lifecycle from database architecture and backend logic to frontend UI and production deployment.

Courses Platform | 04/2025 – 07/2025

Node.js, React | Live

- Constructed the entire backend architecture of a collaborative programming course platform, including database schema, secure user authentication, and data management.
- Integrated the React frontend with the backend APIs to deliver an interactive learning experience.

Clothing Store – NTI | 06/2025 – 07/2025

MEAN Stack | GitHub

- Produced RESTful APIs for a fashion e-commerce platform targeting users aged 18–30, covering dynamic categories, cart and orders, inventory alerts, multi-address checkout, and customer feedback.

Volunteering & Leadership

Graduate Project Team Lead | Obour Institutes | 09/2025 – 06/2026

- Led a 5-member team, distributing tasks and making technical decisions through delivery of the Graduate Project Management Platform (see Projects).
- Coordinated frontend and backend developers to hit the delivery deadline.

PR Core Team Member | IEEE OCS Obour Student Branch | 09/2025 – 06/2026

- Promoted 4 technical events and organized community engagement activities for attendees.

Backend Core Team Member | Google Developer Groups (GDG) Obour Institute | 10/2024 – 07/2025

- Prepared backend workshop content on Node.js and RESTful APIs for 30 students.
- Reviewed backend projects from fellow members.

Problem Solving Coach | ICPC Obour Institute | 09/2024 – 07/2025

- Coached 20 students in competitive programming with C++, covering algorithms and data structures.

Skills

Languages: JavaScript (Node.js), TypeScript, Python, C, C++

Backend: NestJS, Express, Flask, RESTful APIs, JWT Authentication, RBAC, Passport, Mongoose, Multer, Winston

Frontend: React, Angular, TypeScript, HTML, CSS, Tailwind CSS, Material UI, TanStack Query, Axios, Vite

Databases: MongoDB, SQL, MariaDB

DevOps & Cloud: Linux (Ubuntu, Red Hat), Docker, Kubernetes, Ansible, Nginx, Bash, CI/CD, AWS EC2, EBS

Tools: Git, GitHub, Postman

Education

- **Bachelor's Degree in Business Information Systems (BIS)** | Obour Institutes | 10/2022 – 06/2026 | GPA: Excellent
- **Postgraduate Diploma in Computer Science (CS)** | Faculty of Graduate Studies for Statistical Research, Cairo University | 2026 – Present

Certifications

- Red Hat System Administration I (RH124) | maharatech – Red Hat Training | 11/2025 – 12/2025
- Databases | Manara | 08/2025 – 09/2025
- ALX Software Engineering Program | ALX Africa | 05/2023 – 11/2024